# Mehmet Emin Meydanoğlu

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## Experience

### ITU Erdal Aksoy Industrial Robotics and Artificial Intelligence Laboratory
Istanbul, Turkiye

*Undergraduate Research Fellow* — Sep 2026 – Present

Research on generalization in humanoid action policies.

### Hyperever
Istanbul, Turkiye

*Robotics Software Engineer Intern* — Jul – Aug 2026

1. ***Quadruped Blind Rough Terrain Locomotion*** Developed proprioceptive latent-context training methods for quadruped locomotion, inspired by algorithms like RMA, DreamWaQ, and HIM. Tested whether policies could adapt to the environment using latent context inferred from proprioceptive history. Used explicit system identification as the baseline. Transferred a policy trained in Genesis to MuJoCo, where it continued to successfully traverse rough terrain.
2. ***VLM Operator Assistance*** Built a prototype that turned live footage from the robot's front camera into short safety notes for the operator. Uses open-weight edge models like Gemma and Qwen to identify people, obstacles, and terrain conditions and suggest a safe next move. Benchmarked models and frame sampling settings for latency and memory use.

### ITU AUV Team
Istanbul, Turkiye

*Co-Lead of Autonomous Software* — Sep 2025 – Mar 2026

*Autonomous Software Team Fellow* — Sep 2024 – Aug 2026

Building autonomous underwater vehicles (AUVs) for international competitions and academic research. The team placed **third** worldwide at RoboSub 2025, **second** worldwide at RoboSub 2026, and **first** nationally at Teknofest 2026. github.com/itu-auv/auv-software

1. ***Learning-based autonomy*** Experimented with imitation and reinforcement learning for 6-DOF vehicle control and underwater manipulation with robotic grippers, using vision transformer (ViT) image embeddings as policy observations.
2. ***Automated image labeling*** Developed workflows to automate dataset labeling for object detection and segmentation. This work evolved into agent-seg.
3. ***Other classic stuff*** Implemented a custom Pure Pursuit–style controller for trajectory tracking, worked on EKF-based odometry and cascaded PID control, and designed autonomous tasks using state machines.

**Keywords** Reinforcement learning for control, ViT, YOLO, Depth Anything, ROS, visual SLAM

### ITU Racing Team
Istanbul, Turkiye

*Autonomous Software Team Member* — Dec 2023 – Aug 2024

Worked on early-stage autonomous driving software experiments in simulation. Small projects in SLAM, path planning and control for a student formula autonomous vehicle project.

## Projects

**agent-seg** github.com/eminmeydanoglu/agentseg

Tools for agents to label video. Agent will use SAM to track any objects across frames, check and correct labels, and export to any format easily.

**auv-rl** github.com/eminmeydanoglu/auv-rl

Codebase to train reinforcement learning control policies for autonomous underwater vehicles (AUVs).

**self-taught-ViT** github.com/eminmeydanoglu/self-taught-vit

Proof-of-concept self-supervised Vision Transformer trained with masked image reconstruction, proving even a very small ViT can learn semantic representations without any labels.

## Education

**Istanbul Technical University** — **Istanbul, Turkiye**

**B.S. Mathematical Engineering** *(GPA: 3.24/4.00)* — *Sept 2025 – Present*

**B.S. Control and Automation Engineering** *(Transferred)* — *Sept 2023 – Sept 2025*

## Extracurriculars

**Philosophy and science communication:** Produce YouTube videos on philosophy of mind, AI, intelligence, Stoicism, physics, and religion for a general audience. youtube.com/eminmeydanoglu

**Essays:** Write about philosophy, psychology, human nature, meaning, and self-cultivation. meydanoglu.com/writings

**Sports:** Tennis, running, and cycling.